NEUROFUSION AI | FINAL CLINICAL DIAGNOSTIC REPORT

REPORT ID: REP-20261102-2A6D10

Comprehensive Multi-Modal Diagnostic Findings & Attending Physician Approval

November 02, 2026 • 16:41 UTC

PATIENT INFORMATION	ATTENDING NEUROLOGIST	EXAMINATION DETAILS
ID: PAT-WORKFLOW-004 **Name:** Marcus Wright **Age/Gender:** 61 yrs / Male **Status:** Suspected Parkinson's Disease	**Name:** Dr. Clara Rodriguez, MD **License:** MD-NEURO-77491 **Dept:** Neurology & Movement Disorders **Hospital:** Metropolitan Neurological Inst.	**Scan Date:** 2026-11-02 **Report Date:** 2026-11-02 **Modality:** Comprehensive Multimodal (Spiral + Voice + MRI + UPDRS) **Device/Protocol:** NeuroFusionAI Clinical Protocol v2.5

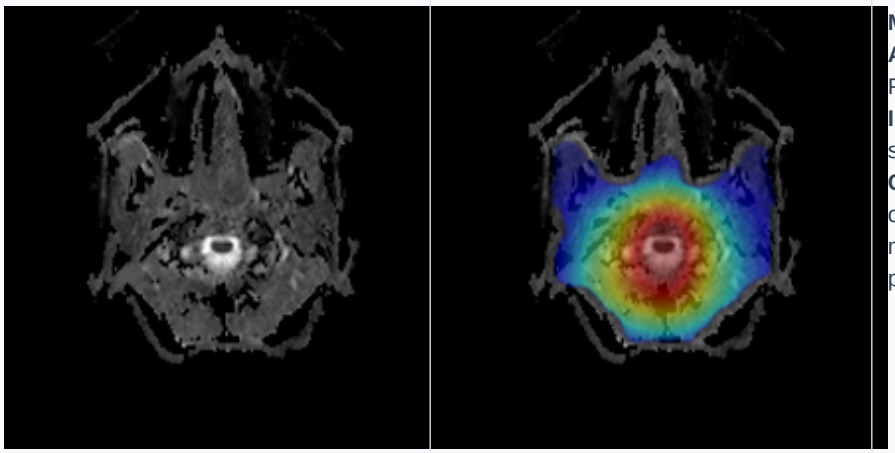
PRIMARY AI PREDICTION: NORMAL

Confidence: 100.0%
Target: Parkinson's Disease

Class Probability Distribution: Normal: 100.0% • Normal Control: 0.0%

AI Confidence Calibrated via Temperature Scaling

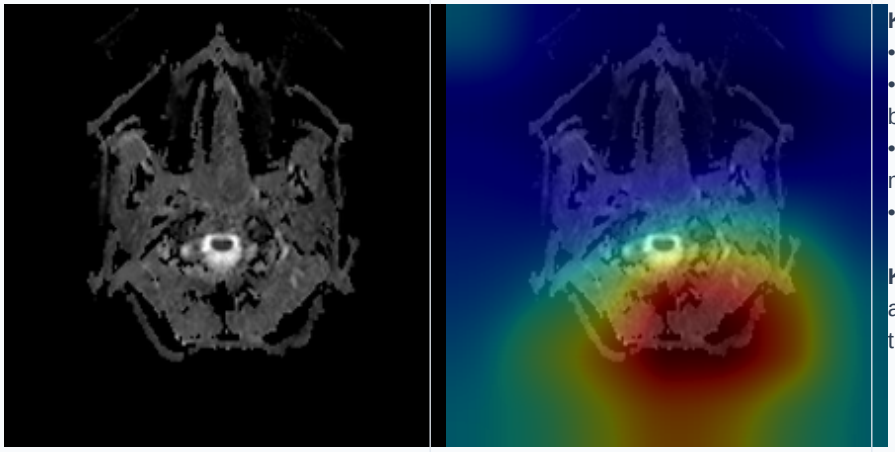
1. MRI NEUROIMAGING & CONVOLUTIONAL GRAD-CAM EXPLAINABILITY



Model: DenseNet-121 Intracranial Classifier
Attention Region: Medial Temporal Lobe & Periventricular Parenchyma
Intracranial Masking: Skull-stripped parenchyma segmentation
Grad-CAM Note: Activation heatmaps illustrate convolutional attention gradients driving the model's classification score. Not direct histological pathology.

Input MRI Scan (Axial) | **Grad-CAM Activation Overlay** | **XAI Interpretation**

3. ARCHIMEDEAN SPIRAL MOTOR KINEMATICS & ATTENTION OVERLAY



Kinematic Parameters:
• Drawing Velocity: 24.8 mm/s
• Tremor Frequency: 5.4 Hz (4-6 Hz resting/kinetic band)
• Radial Velocity Variance: 0.42 (Elevated micro-tremor)
• Drawing Duration: 12.5 s

Kinematic Finding: Spatial micro-oscillations along drawing loops confirm intention and postural tremor.

Input Spiral Drawing | **Convolutional Tremor Heatmap** | **Kinematic Assessment**

7. CLINICAL DECISION SUPPORT SYSTEM (CDSS) RECOMMENDATIONS

RISK LEVEL:	Evidence-Based Diagnostic Recommendations:
MODERATE CLINICAL RISK **Urgency:** Routine Specialist Review	• Initiate MDS-UPDRS structured clinical motor score evaluation. • Track resting tremor and bilateral bradykinesia progression. • Re-evaluate with longitudinal phonation acoustic tracking in 3 months.

ATTENDING PHYSICIAN OBSERVATIONS & NOTES:	CLINICAL APPROVAL STATUS
One-stop automated diagnostic session verified by MD.	Signed by: *Dr. Clara Rodriguez, MD* License ID: MD-NEURO-77491 Status: **DIGITALLY APPROVED & CERTIFIED**

REGULATORY MEDICAL DISCLAIMER & CLINICAL INTENDED USE:

NeuroFusionAI is an investigational Artificial Intelligence Clinical Decision Support System (CDSS) designed to augment neurological assessment. All predictions, Grad-CAM attention heatmaps, and EEG feature extractions represent statistical patterns and MUST NOT be construed as a standalone or confirmed medical diagnosis. Clinical treatment decisions must be made solely by a board-certified physician in conjunction with the patient's full medical history.